Water, chemical admixtures and mineral admixtures

Notes from the lecture slides "Module 3 – Admixtures (3a Quality of water, 3b Chemical admixtures, 3c Mineral admixtures)"

In this module

1. Part A: Quality of water
2. Part B: Chemical admixtures
3. Part C: Mineral admixtures (supplementary cementing materials)

1 Part A: Quality of water

1.1 Why water matters

Water has two jobs in concrete:

- **Mixing**: water reacts with cement (hydration), which produces the strength and durability of concrete. It also makes the mix workable.
- **Curing**: keeping concrete moist and at a suitable temperature in the early stages of hardening, so that hydration continues and strength develops. Curing is the **control of temperature and of moisture movement into and out of the concrete**; it decides both strength and durability.

Impurities in water can affect the **setting of cement**, reduce the **strength of concrete**, **stain** the surface, or cause **corrosion of reinforcement**.

1.2 Mixing water

- Most specifications say mixing water should be **fit for drinking** (potable). Drinking water rarely has more than **2000 ppm** of dissolved solids, and usually less than 1000 ppm.

- If the **silt content is above 2000 ppm**, let the water stand in a **settling basin** before use.
- **Wash water from truck mixers** is satisfactory, because the solids in it are concrete ingredients anyway (cement paste, fine aggregate). ASTM C94 allows it, provided it does not mix different cements or different admixtures.

1.3 Is drinking water always suitable?

The "fit for drinking" rule is **not absolute**, in either direction:

- Drinking water may be **unsuitable** if it has a **high concentration of sodium or potassium**, because these alkalis can cause **alkali–aggregate reaction**.
- Water that is not drinkable may still be fine. A practical rule: any water with **pH 6.0 to 8.0** that does **not taste saline or brackish** can be used.
- **Colour or smell** does not necessarily mean harmful substances are present.
- Water containing **humic or other organic acids** may harm the hardening of concrete; avoid it or test it. **Highly alkaline water** should also be tested.
- **Algae** in mixing water **entrain air**, which lowers strength.

1.4 Harmful substances

- **Brackish water** contains chlorides and sulfates. Limits in the slides: **chloride below 500 ppm** and **SO_3 below 1000 ppm**. Higher amounts cause **chloride attack and sulfate attack**, and chlorides cause **corrosion of reinforcement**.
- **Alkali carbonates and bicarbonates** should be below **1000 ppm** (they affect setting time).

For reference, **IS 456:2000** gives these limits for mixing water:

Impurity	Maximum
Organic solids	200 mg/l
Inorganic solids	3000 mg/l
Sulfates (as SO_3)	400 mg/l
Chlorides (as Cl)	2000 mg/l for plain concrete; **500 mg/l for reinforced concrete**
Suspended matter	2000 mg/l
pH	Not less than 6

1.5 Sea water as mixing water

- Sea water has a total salinity of about **3.5%**: about **78% of the dissolved solids is NaCl** and **15% is $MgCl_2$ and $MgSO_4$**.
- It gives slightly **higher early strength but lower long-term strength**; the loss is usually **not more than 15%**, which may be tolerable.
- The **initial setting time** must be checked (it can be shortened).
- It can cause **dampness and efflorescence** (white salt deposits).
- Its biggest danger is **corrosion of reinforcement**, especially in **tropical countries**, in structures **exposed to humid air**, where **cover is inadequate** or the concrete is **not dense**.
- If reinforced concrete is **permanently under water**, there is little ill effect (not enough oxygen for corrosion).
- **In general, sea water should not be used**, and never for reinforced or prestressed concrete.

1.6 Curing water

- **Iron or organic matter** in curing water can **stain** the concrete. Discolouration does not matter structurally, so water suitable for mixing, or even slightly inferior, is acceptable for curing.
- What matters is that curing water must be **free from substances that attack hardened concrete**. Concrete is attacked by:
  - water containing **free CO_2** (it is acidic), and

- **flowing pure water** (from melting ice or condensation, with little CO_2), which **dissolves $Ca(OH)_2$** and erodes the surface.

1.7 Cover concrete and core concrete

The **cover concrete** ("cover-crete") is the outer layer between the surface and the reinforcement; the **core** ("heart-crete") is the concrete inside. Harmful agents (water, chlorides, CO_2, sulfates) attack from outside, so the **cover concrete must be of high quality and low permeability** to protect the steel. The cover is also the part that dries out first, so it suffers most from poor curing. **Adequate curing is essential for durable concrete.**

1.8 Testing whether water is suitable

- Compare the **setting time of cement** and the **strength of mortar cubes** made with the water in question against those made with **distilled water** (or ordinary drinking water; there is no appreciable difference between them).
- **BS 3148** allows a **10% tolerance** in strength. (IS 456 similarly requires the 28-day strength to be at least **90%** of the control, and the initial setting time to be at least 30 minutes and within **±30 minutes** of the control.)
- For curing water, a **performance test** with repeated **wetting and evaporation** shows whether impurities will stain the surface.

2 Part B: Chemical admixtures

2.1 What an admixture is

An **admixture** is an extra ingredient, besides cement, other cementitious materials, water and aggregate, added to concrete to change its **fresh (plastic) or hardened properties**, for example **workability, strength or durability**.

ACI 116R definition: "a material other than water, aggregates, hydraulic cement, and fibre reinforcement, used as an ingredient of concrete or mortar, and added to the batch immediately before or during its mixing."

For uniform distribution, the admixture is usually **mixed with the mixing water**.

2.2 Why use chemical admixtures?

- To achieve or improve **specific properties** in the fresh and hardened states.
- To keep **quality** during mixing, transporting, placing and curing in **adverse weather** (very hot or cold).
- To deal with **emergencies** during site work (e.g. delays, breakdowns).
- To **reduce overall cost** for the same properties and durability.

2.3 Classification

Group	Types
Set-controlling	Accelerators; Retarders
Water reducers	Normal (plasticizers); Mid-range; High-range (superplasticizers)
Air entrainers	–
Special admixtures	Viscosity modifiers; Corrosion inhibitors; Shrinkage-reducing admixtures; SBR latexes; others (antifreeze, waterproofing, ASR-mitigating)

2.4 Set-controlling admixtures

Set controllers are organic or inorganic chemicals that **interfere with cement hydration**: they either **speed up** (accelerators) or **slow down** (retarders) the dissolution of the cement compounds that hydration depends on.

	Accelerators	Retarders
Uses	Earlier finishing of slabs; high early strength; **early removal of formwork**; **cold-weather concreting**	**Hot-weather concreting**; preventing **cold joints** between successive lifts; **long hauls** (ready-mix); keeping concrete workable for longer
Chemicals	**Chloride**: $CaCl_2$, NaCl. **Non-chloride inorganic**: nitrates and nitrites of Ca and Na, thiocyanates, thiosulphates. **Non-chloride organic**: amines (triethanolamine TEA, diethanolamine DEA), carboxylic acids (Ca salts of formic and acetic acid), formaldehyde	**Organic**: lignosulphonates, hydroxycarboxylic acids (citric, gluconic), carbohydrates (corn syrup, dextrin): the same chemicals as normal water reducers. **Inorganic**: borates, phosphates, Zn and Cu compounds (rarely used: costly, low solubility). **Extended-set**: phosphonates and other phosphorus-containing organic acids and salts, gluconic acid

- **$CaCl_2$** at **1–3% by weight of cement** is the most effective accelerator, but chlorides **promote corrosion of steel**, so chloride accelerators are avoided in reinforced and prestressed concrete; non-chloride accelerators are used instead.
- **Extended-set retarders** can hold concrete plastic for many hours. They are used to **stabilize returned plastic concrete** (so it can be reused) and for **very long haul times**.

How set controllers work (Joisel's theory)

They act during the early stage of hydration and affect only **dissolution**. Think of hydrating cement as a source of **cations (Ca^{2+})** and **anions (silicate and aluminate)**.

1. An **accelerator** must speed up dissolution of both cations and anions. Among the anions, it must speed up the one that **dissolves slowest: silicate**.
2. A **retarder** slows down dissolution of cations or anions, mainly the anion that **dissolves fastest in the early period: aluminate**.
3. **Monovalent cations (K^+, Na^+)** reduce the solubility of Ca but increase that of silicate and aluminate. At **low** concentration the first effect dominates (retardation); at **high** concentration the second dominates (acceleration).

4. **Monovalent anions (Cl^-, NO_3^-)** and SO_4^{2-} reduce the solubility of silicate and aluminate and increase that of Ca^{2+}.

Practical issues

- Add the admixture **soon after cement and water come into contact**.
- **Watch the dosage**: the **same chemical can act as an accelerator or a retarder** depending on its concentration.

The slides show five types of behaviour of initial setting time against admixture concentration:

Type	Chemicals	Behaviour (from the graph)
I	Gypsum	Setting time rises with dosage, then levels off
II	Calcium chloride, calcium nitrate	Retard at very low dosage, accelerate at higher dosage
III	Potassium and sodium carbonate, sodium silicate	Slight retardation at low dosage, then acceleration
IV	Gluconates, lignosulphonates and sugars, sodium salts of carboxylic acids, Zn and Pb salts	Strong retarders: setting time rises steeply
V	Salts of formic acid, triethanolamine (TEA)	Setting time falls as dosage increases

At small concentrations, salts of a weak base and strong acid ($CaCl_2$) or of a strong base and weak acid (K_2CO_3) retard the dissolution of calcium and aluminate more than they accelerate silicate, so they retard; at higher concentrations they accelerate.

2.5 Water reducers

A water reducer lets you do one of three things:

1. **Same workability, less water** → lower w/c → **higher strength and durability**.
2. **Same w/c and strength, more workability**.
3. **Same w/c, strength and workability, less cement** → cheaper, less heat and shrinkage.

Type	Water reduction	Typical dosage
Normal (plasticizers)	5–8%	0.3–0.5% by weight of cement
Mid-range	8–15%	–
High-range (superplasticizers)	15–25% (PCE up to ~40%)	0.7–1.0% by weight of cement

The slides' graph of flow-table spread against water content shows the two curves (with and without SP): at the same water content the SP mix flows much more (**increased workability**), and for the same flow it needs much less water (**increased strength**).

Normal water reducers

- **Lignosulphonate salts** (sodium salts of sulphonated lignin, a by-product of paper making)
- **Hydroxycarboxylic acids**: citric, gluconic acid
- **Carbohydrates**: corn syrup, dextrin
- **Overdosing** causes **excessive retardation, bleeding and air entrainment**.

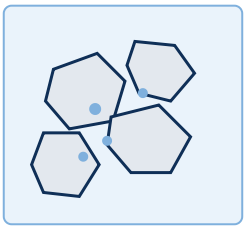
Superplasticizers (high-range water reducers)

- Reduce water by about **25%**; used mainly to make **high-strength concrete** at a given workability.
- Strength of about **100 MPa** is possible at **w/c = 0.28**.
- Even a **small dose (about 0.1%)** greatly increases the flow of a cement paste, more than adding 10% extra water.

1st generation	2nd generation	3rd generation
Lignosulphonates at high dosage	**Sulphonated melamine formaldehyde (SMF)**; **sulphonated naphthalene formaldehyde (SNF)**	**Polycarboxylates (PCE)**, polyacrylates, monovinyl alcohols

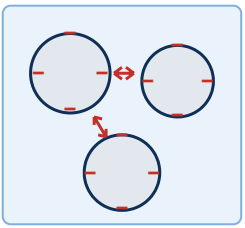
How water reducers work

Without admixture



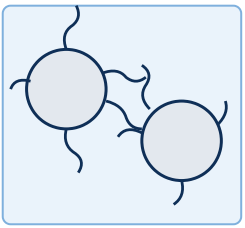
grains flocculate and trap water (blue) inside flocs

SNF / SMF / lignosulphonate



adsorbed molecules give the grains like (negative) charges: **electrostatic repulsion**

PCE (polycarboxylate)



long side chains stick out and physically keep grains apart: **steric hindrance**

Either way the grains are dispersed, the trapped water is released, and the mix flows more for the same water content.

PCE also keeps working longer (better slump retention) and allows up to ~40% water reduction.

Without an admixture, cement grains clump together (flocculate) and trap water. Water reducers disperse them.

Electrostatic repulsion (lignosulphonates, SNF, SMF, sugar-refined lignosulphonates):

- Cement grains carry surface charges and tend to **flocculate**, trapping water inside the flocs.
- The admixture molecule has an **anionic polar group** on a hydrocarbon chain. It is **adsorbed on the cement grain surface**.
- All grains then carry a **like (negative) charge**, so they **repel each other**. (The slides describe this in terms of the **zeta potential**, the electrical potential at the grain surface, which the admixture changes.)
- The polar ends also make the cement grains **hydrophilic** and lower the surface tension of water.
- Result: the grains disperse, the **trapped water is released**, and the mix becomes more fluid.

Steric hindrance (new-generation PCE):

- PCE polymers have a **backbone** that adsorbs on the grain and **long graft (side) chains** that stick out into the water.
- The side chains **physically stop the grains coming close** to each other, in addition to some electrostatic repulsion.
- Types: **polycarboxylic ether (PCE)**, carboxylic acrylic acid with acrylic ester, cross-linked acrylic polymers.
- **Advantages:** water reduction **up to 40%**, **good slump retention**, effective at **low w/c**.

Effect of PCE structure:

Desired performance	Backbone length	Side chain length	Number of side chains
Low dispersion, short retention	Long	Short	Many
High dispersion	Short	Long	Few
Long slump retention	Shorter	Long	Many

Common problems with water reducers

Type	Issues	Best suited to
Unrefined lignosulphonates	Variable properties, entrain air, inconsistent performance	–
Melamine formaldehyde (SMF)	**No slump retention**, costly	**Cold weather and precast**; not for ready-mix with long hauls
Naphthalene formaldehyde (SNF)	Not compatible with PC (polycarboxylate) admixtures	**Hot tropical climates**; low cost
Polymer-based (PCE)	Limited experience, expensive	High-performance, self-compacting concrete

2.6 Air-entraining admixtures

The slides list air entrainers in the classification without a separate slide. What you need to know:

- They are surfactants (e.g. **vinsol resin**, fatty acid salts, sulphonated hydrocarbons) that stabilize **millions of tiny, separate air bubbles** (about 10 µm to 1 mm) in the paste.
- Main purpose: **resistance to freezing and thawing**. When water in the pores freezes and expands, the bubbles give it space, so the pressure does not crack the paste. Typical air content is about **4–6%**.
- They also **improve workability and cohesion** and **reduce bleeding and segregation**.
- Disadvantage: **strength falls by about 5% for each 1% of air**, so the w/c ratio is usually reduced to compensate.

2.7 Viscosity-modifying admixtures (VMA)

- Give **stability** to very flowable concrete (e.g. self-compacting concrete) that might otherwise **segregate**, and **reduce bleeding, segregation and settlement**.
- Prevent **wash-out** of concrete placed **under water**; then called **anti-washout admixtures**.
- They are **long-chain water-soluble polysaccharides** that increase the paste's water-retention capacity: cellulose ether derivatives and microbial polysaccharides (**welan gum**, xanthan gum, diutan gum), alginates, hydroxypropyl methyl cellulose, hydroxypropyl starch.
- **Dosage:** about **0.03–0.08% by weight of cement**.

How VMAs work:

1. **Adsorption**: the polymer molecules attach to water molecules, fixing part of the mixing water and swelling, which raises viscosity.
2. **Association**: neighbouring polymer chains attract each other, forming a **viscous gel** that further blocks water movement.
3. **Intertwining**: at low shear rates the chains **entangle**, raising viscosity; at high shear rates (during pumping or mixing) they **disentangle and align with the flow**, so viscosity drops. This is **shear thinning**: the concrete is stable when still but flows when moved.

2.8 Shrinkage-reducing admixtures (SRA)

- When concrete dries, water leaves the capillary pores (mainly **2.5–50 nm**). **Menisci** form in the pores because of the **surface tension** of water, and they pull **inward on the pore walls**, causing **drying shrinkage**.
- SRAs (e.g. **polyoxyalkylene alkyl ether**) **lower the surface tension** of the pore water, so the inward pull and the shrinkage are reduced.
- High dosage: about **2–4% by weight of cement**.

2.9 Corrosion inhibitors

They protect steel by one or more of:

- **passivating** the steel (oxidizing or non-oxidizing passivators),
- **scavenging oxygen**,
- forming a **protective film** on the steel (adsorption), and
- a **cathodic effect**, e.g. making the paste **hydrophobic** so less water reaches the cathode.

Examples: **inorganic**: **calcium nitrite**; **organic**: amines, esters, alkanolamines. Used at high dosage (about **2% by weight of cement**), so **very expensive**.

Some products are **two-part**: the **amines** form a film on the steel, while the **esters** make the paste hydrophobic, reducing the water available for the cathodic reaction.

2.10 Other special admixtures

- **Antifreeze admixtures** lower the freezing point of water: NH_4OH, calcium and sodium nitrates and nitrites, $CaCl_2$, K_2CO_3, glycols.
- **Waterproofing admixtures**: organic compounds that **adsorb on pore walls and make them hydrophobic**, so dry concrete is hard to re-wet: oleic acid, wax emulsions, calcium and aluminium stearate.
- **ASR-mitigating admixtures**: **lithium** compounds (LiOH, lithium nitrate LiNO3) bind reactive silica into non-expansive products, but they are very costly.
- **SBR latex** (styrene–butadiene rubber) improves bond and reduces permeability; used in repair mortars.

3 Part C: Mineral admixtures (supplementary cementing materials)

3.1 Why mineral admixtures?

Concrete is the **second most consumed material** in the world after water. Good concrete needs:

Property	Mainly controlled by
Workability	Water content, chemical admixtures
Compressive strength	**Pore volume**
Durability	**Pore connectivity**
Dimensional stability	Efficient packing of particles, water content

The weak part of concrete is in the hydration products:

- **C-S-H** (main binder, poorly crystalline gel): about 60–65%
- **$Ca(OH)_2$** (hexagonal crystals): about 15–20%
- **Calcium sulphoaluminates** (AF_m, AF_t): about 10–15%

Problems: **$Ca(OH)_2$ is the weakest product and is easily attacked chemically**; and **pores** are not filled efficiently, so they are interconnected, especially in the **interfacial transition zone** around aggregates.

Solution: convert $Ca(OH)_2$ into C-S-H, and fill the pores with **fine mineral particles**. This is what mineral admixtures, also called **supplementary cementing materials (SCMs)**, do.

Benefits:

- **Serviceability**: better long-term durability; less cracking (lower heat).
- **Sustainability**: uses **industrial wastes and by-products**; less clinker (cement production causes about **6–7% of total greenhouse-gas emissions**).
- **Economy**: longer service life; corrosion repairs cost about **5% of GDP** in developed nations.

3.2 Role of SCMs and the pozzolanic reaction

Pozzolan (ASTM C595 definition): a **siliceous or siliceous and aluminous material** which by itself has **little or no cementitious value**, but which, **in finely divided form and in the presence of moisture**, reacts with **calcium hydroxide** at ordinary temperatures to form compounds with **cementitious properties**.

SCMs act in two ways:

1. **Chemically**: some are **cementitious** (react with water themselves, e.g. slag), most are **pozzolanic** (react with the CH from cement hydration). Either way, they produce **additional C-S-H**.
2. **Physically (filler effect)**: their fine particles **fill small pores**, giving a **denser structure** with lower porosity and **lower permeability**.

$$Ca(OH)_2 + \text{reactive } SiO_2 \text{ (or } Al_2O_3) + H_2O \rightarrow \text{C-S-H (or C-A-H)}$$

The pozzolanic reaction is:

- **lime-consuming** (removes the weak, soluble CH),
- **pore-refining** (fills capillary pores with C-S-H),
- **interface-refining** (strengthens the transition zone around aggregates),
- **slow**, and
- **accelerated by alkalis and gypsum**.

Pozzolanic activity index (PAI)

$$\text{PAI (\%)} = \frac{\text{strength of cement + pozzolan mix}}{\text{strength of plain cement mix}} \times 100$$

A higher PAI means a more reactive pozzolan.

3.3 Classification of SCMs

Class	Examples
Cementitious	(Portland cement itself)
Highly pozzolanic	**Silica fume**; **rice husk ash** (controlled burning)
Normal pozzolans	**Class F fly ash**
Cementitious and pozzolanic	**GGBFS**; **Class C fly ash**
Weak pozzolans	Slowly cooled slag, bottom ash, field-burnt rice husk ash

On a $CaO–SiO_2–Al_2O_3$ diagram, silica fume sits near pure SiO_2, natural pozzolans and Class F fly ash are high in silica and alumina, and slag and Class C fly ash lie between them and Portland cement (which is rich in CaO).

3.4 Fly ash

Source and collection

- A **by-product of burning pulverized coal in thermal power plants**. Its quality depends on the **coal**:
  - anthracite (90–95% C): hard coal, needs high temperature to burn;
  - semi-bituminous: good heating value, smokeless flame;
  - bituminous (80–90% C): soft coal;
  - sub-bituminous (70–80% C); and
  - lignite (brown coal).
- Low-quality coals contain clay, shale, quartz, carbonates and sulfides, which decide the composition of the ash.
- **75–80% of the ash flies out with the flue gas**, hence "fly ash". The rest is **bottom ash** (can be processed as aggregate but is generally not used in concrete).
- Fly ash is collected in **electrostatic precipitators**: smoke particles pick up a negative charge, are attracted to positively charged collecting plates, and are knocked off into hoppers. Much fly ash is still **dumped in ponds as slurry**.

Uses: mineral admixture; filler; **synthetic (lightweight) aggregate**, made by sintering or by agglomeration with lime or cement (expensive).

Class C and Class F (ASTM C618)

	Class C (high calcium)	**Class F** (low calcium)
Source coal	Lignite, sub-bituminous	Bituminous, anthracite
Behaviour	**Both cementitious and pozzolanic**	**Normal pozzolan** only
Particles > 45 µm	10–15%	15–20%
Blaine fineness	300–400 m^2/kg	200–300 m^2/kg
Average size	Below 20 µm	About 20 µm
Shape	Mostly solid spheres, smooth	Solid spheres, smooth

Effect on fresh concrete

- **Setting time increases** (slower reaction).
- **Workability and flow increase**: the **spherical particles act like ball bearings**.
- **Bleeding and segregation are reduced** in well-proportioned mixes.
- **Paste volume increases** when cement is replaced **by mass**, because fly ash is lighter than cement (specific gravity about 2.2 vs 3.15).

Effect on hardened concrete

- **Strength gain is slower**, but **ultimate strength (after 28 days) is usually higher** with proper curing.
- **Creep and shrinkage are typically higher**, because of the larger paste volume (with mass replacement).
- Effect on **sulfate resistance**: results are **inconclusive**.
- **Alkali–aggregate expansion is reduced**, because cement is diluted (fewer alkalis available).
- **More air-entraining admixture** is needed for the same air content.

Problems with fly ash

- **Loss on ignition (LOI)** indicates **unburnt carbon**; too much carbon **absorbs air-entraining agent**, giving poor air-void systems.

- Limits are also placed on **SO_3, MgO, alkali and moisture content**.
- **Transport cost**: beyond about **40–50 km** from the power plant, using fly ash becomes **uneconomical**.
- Poor handling and transport facilities: **little control over the quality** received from the plant.

Special applications

- **High-strength concrete** (as extra cementitious material)
- **Roller-compacted concrete** (helps bond between layers)
- **Controlled low-strength materials (CLSM)**: flowable mortars used as **backfill**
- **Synthetic aggregate** and **bricks**

3.5 Silica fume

Source and production

- A **by-product of the silicon and ferrosilicon industry**. Quartz is reduced with carbon (coke, coal, wood chips) in an **electric arc (smelting) furnace at about 2000 °C**:

$$SiO_2 + 2C \rightarrow Si + 2CO$$

- Some SiO vapour escapes with the off-gas, oxidizes and condenses into **extremely fine spheres of amorphous silica**. The composition depends on the alloy being made (silicon metal gives purer fume).
- The fume is carried from the furnace by large pipes, cooled, passed through **cyclones** (to remove oversize particles), and captured in a **baghouse** (fabric filter).

Forms available

Form	Notes
As-produced powder	Very low specific gravity (~2.2) and bulk density, so it is **bulky and hard to handle and transport**
Densified	Compacted by pressure, which **flocculates** the particles; needs an efficient **superplasticizer** to break them up and disperse them. Undispersed lumps act like aggregate (no pozzolanic activity) and reactive silica lumps can cause **ASR**
Slurry	50% water + 47% silica fume + 3% chemical agent (keeps particles suspended, prevents gelling). Efficient to dispense and saves storage space, but can **gel in cold climates**

Colour: premium white (purer, about 85–95% SiO_2 or more) and standard grey; **whiter means purer**.

Physical properties

Property	Value
Particle size	**Less than 1 µm** (about 100 times finer than cement)
Bulk density	As produced 130–430; densified 480–720; slurry 1320–1440 kg/m^3
Specific gravity	**2.2**
Surface area (BET)	**13,000–30,000 m^2/kg**
Typical replacement	**5–15%** of cement (commonly 5–10%)
Cost	About **10 times** that of Portland cement

Benefits: a strong **pozzolanic reaction** producing extra C-S-H, plus a strong **filler effect** from the very fine particles.

Effect on fresh concrete

- Its high fineness **increases water demand**, so it is **always used with a superplasticizer**.
- The mix becomes **sticky and cohesive**, and is **prone to slump loss**. Because of the cohesiveness, a **higher slump** is needed for placing.

- **Bleeding is drastically reduced**; most silica fume mixes do not bleed at all. In dry, windy conditions, if evaporation is faster than bleeding, **plastic shrinkage cracking** can occur, so early curing is essential.

Effect on hardened concrete

- **Pore size refinement** and much **lower permeability**.
- **Higher compressive and flexural strengths**.
- **Higher elastic modulus** (about 115% of plain concrete): stiffer, and more brittle.
- **Creep and shrinkage increase** at high replacement levels (10–15%) because of the larger paste volume.
- **More air-entraining agent** is needed; freeze–thaw resistance is slightly lower, but damage is limited by the very low permeability.
- **Better resistance to chemical attack** (because of low permeability and less CH), except to **ammonium sulfate and magnesium sulfate**.
- **ASR expansion is reduced**.
- **Corrosion rate is reduced**, because (1) low permeability limits **moisture and oxygen at the cathodic sites**, and (2) the **high electrical resistivity** makes current flow difficult.
- **Carbonation depth is generally lower**.
- **Very good abrasion and erosion resistance**.
- **Fire performance is poor** (the dense paste cannot release steam, so it spalls).

Summary: **more durable concrete, but not good against fire**.

3.6 Ground granulated blast-furnace slag (GGBFS)

Source

- Mainly **silicates and aluminosilicates of calcium**.
- A by-product of **iron making**: iron ore, coke and limestone (flux) are fed into a blast furnace. Impurities in the ore combine with lime and **float on top** of the molten iron as slag; the heavier molten iron stays at the bottom.

- Portland blast-furnace slag cement was first made in **1892**; slag is now used in almost every country.

Cooling decides reactivity

The **faster the molten slag is cooled, the more glassy and reactive it is**. In increasing order of reactivity:

1. **Slow cooling in air** (crystalline, least reactive)
2. **Rapid cooling by water spray**
3. **Quenching** (dipping in water): glassy, most reactive

Type of slag	Reactivity	Use
Air-cooled	Low	Aggregate; strong and tough, ideal for **railway ballast**
Expanded (foamed)	Low	Foamed with air; **lightweight aggregate**, thermal insulation
Granulated	**High**	Quenched, then **ground to cement fineness** → **GGBFS**
Pelletized	Similar to GGBFS	Complex process; less used

What governs GGBFS performance

- Chemical composition of the slag
- Alkali concentration of the reacting system
- **Glass (reactive SiO_2) content**
- Fineness of the GGBFS and of the cement
- Temperature during early hydration

GGBFS is off-white (lighter than cement).

Activation

Slag reacts only slowly with water on its own, so an **activator** is needed:

- **Alkali activation** (NaOH, Na_2CO_3, sodium silicate): products are C-S-H, C_4AH_{13} and **C_2ASH_8 (gehlenite hydrate)**.

- **Sulfate activation** (gypsum, hemihydrate, anhydrite, phosphogypsum): products are C-S-H, **ettringite** and **aluminium hydroxide (AH_3)**.
- **Mixed activation**: both alkali and sulfate present, as in a **Portland cement–slag blend** (the cement supplies both).

Effect on concrete

- Apart from **delaying initial set and early strength gain**, slag does **not change fresh properties much**. Slag particles are angular (ground, not round), so at high replacement a **plasticizer** may be needed.
- **Early strength gain is slower**; the early rate is **inversely proportional to slag content**.
- **Ultimate strength and durability improve**.
- Slag is the **material of choice in marine environments**: its high alumina **binds chlorides**, reducing the risk of corrosion, and it lowers permeability.

3.7 Rice husk ash (RHA)

- A **highly reactive pozzolan** made by **controlled burning (calcination)** of rice husk.
- **Field-burnt** husk ash is largely **crystalline**, so it is a **weak pozzolan** with little activity; **controlled burning** is needed to keep the silica amorphous.
- Often contains **unburnt carbon**, which interferes with air entrainment.
- Particles below **45 µm**, with a very high internal surface area (about **60,000 m^2/kg**) because the particles are **cellular** (porous).
- **Over 90% reactive silica**.

3.8 Metakaolin (MK)

- Made by **calcining kaolinite clay at 740–840 °C**. The clay loses its bound water and its crystal structure, becoming amorphous and reactive. Burning must stay in this range: **above about 1000 °C the clay recrystallizes** and loses reactivity.
- General formula **AS_2** ($Al_2O_3 \cdot 2SiO_2$).
- Reacts with CH to give extra C-S-H (and calcium aluminate hydrate):

$$AS_2 + 6\,CH + 9\,H \rightarrow C_4AH_{13} + 2\,C\text{-}S\text{-}H$$

- The C-S-H formed is **aluminous**, with C/S from **0.83** (crystalline forms) to **above 1.5** (amorphous forms).
- The amount and rate depend on the **mineralogy of the kaolin**.
- **Performance comparable to silica fume**, but MK is **manufactured, not a by-product**, so it is **expensive** and less marketable than silica fume.

3.9 Calcined clay–limestone blends (LC3)

- **Limestone reacts with alumina** to form **mono-carboaluminate** (and hemi-carboaluminate), which fills space and adds strength.
- In OPC, up to **5% limestone** is beneficial because of this reaction.
- When alumina is readily available from **calcined clay**, **much higher replacement** is possible. This is the basis of **LC3 (limestone calcined clay cement)**, typically about 50% clinker, 30% calcined clay, 15% limestone and 5% gypsum.
- Notes from class: the higher the replacement, the lower the strength, so high-replacement blends suit **lower-strength concrete such as residential buildings**.

3.10 Rate of pozzolanic reaction

Typical time for the reaction to become significant (t_{eff}):

SCM	t_{eff}
Rice husk ash	1 day
Silica fume	1–2 days
Class C fly ash	3–7 days
GGBFS	3–7 days
Class F fly ash	**14–28 days**

So concrete with SCMs, especially Class F fly ash, needs **longer moist curing**, and its strength is often specified at **56 days** instead of 28.

Quick revision

- Mixing water: potable usually fine; pH 6–8; Cl < 500 ppm, SO_3 < 1000 ppm (slides); IS 456: Cl ≤ 500 mg/l for RCC, SO_3 ≤ 400 mg/l. Test: setting time and ≥ 90% strength vs distilled water.
- Sea water: 3.5% salts; higher early, lower later strength (≤ 15% loss); efflorescence; corrosion risk → avoid, never in RCC.
- Curing water: staining harmless; avoid free CO_2 and flowing pure water.
- Accelerators: $CaCl_2$ (1–3%, not in RCC), nitrates, TEA, formates → cold weather, early stripping. Retarders: lignosulphonates, sugars, citric/gluconic acid, phosphates → hot weather, long haul, cold joints.
- Joisel: accelerator boosts silicate dissolution; retarder blocks aluminate dissolution. Same chemical can do either depending on dose.
- Water reducers: normal 5–8% (0.3–0.5%), mid 8–15%, SP 15–25% (0.7–1.0%), PCE up to 40%.
- SNF/SMF: electrostatic repulsion. PCE: steric hindrance, better slump retention.
- AEA: 4–6% air, freeze–thaw resistance, ~5% strength loss per 1% air. VMA 0.03–0.08%, shear thinning. SRA 2–4%, lowers surface tension. Corrosion inhibitor: calcium nitrite ~2%.
- Pozzolanic reaction: CH + S + H → C-S-H; lime-consuming, pore- and interface-refining, slow. PAI = strength ratio × 100.
- Fly ash: Class C (high Ca, cementitious + pozzolanic) vs Class F (low Ca, pozzolanic); ball-bearing effect; slow early strength; ASR reduced; LOI = unburnt carbon.
- Silica fume: < 1 µm, SG 2.2, 13–30 thousand m^2/kg, 5–15%, needs SP, no bleeding, high strength and E, poor fire resistance.
- GGBFS: quenched slag, needs activator, slow early strength, best for marine work.
- RHA: controlled burning, > 90% reactive silica. MK: kaolin at 740–840 °C, AS_2. LC3: clinker + calcined clay + limestone.
- t_{eff}: RHA 1 d < SF 1–2 d < GGBFS, Class C 3–7 d < Class F 14–28 d.

Questions to practise

1. What are the requirements of water for mixing and curing concrete? How is the suitability of water tested? (8 marks)
2. Can sea water be used for making concrete? Discuss. (5 marks)
3. Define an admixture. Classify chemical admixtures and give two examples of each. (6 marks)
4. Compare accelerators and retarders: chemicals used, applications and precautions. (8 marks)
5. Explain the mechanism of set-controlling admixtures according to Joisel. (5 marks)
6. What are superplasticizers? Explain how SNF and PCE superplasticizers disperse cement particles, with sketches. (10 marks)
7. Write short notes on: (a) viscosity-modifying admixtures, (b) shrinkage-reducing admixtures, (c) corrosion inhibitors, (d) air-entraining admixtures. (4 × 4 marks)
8. What is a pozzolan? Write the pozzolanic reaction and explain how mineral admixtures improve concrete. (6 marks)
9. Compare Class C and Class F fly ash. How does fly ash affect fresh and hardened concrete? (10 marks)
10. Describe the production, properties and effects of silica fume on concrete. (10 marks)
11. What is GGBFS? How does the method of cooling affect its reactivity? How is slag hydration activated? (8 marks)
12. Write short notes on rice husk ash, metakaolin and LC3. (9 marks)